End-To-End Retail Retention Analysis

Where does our revenue come from, and how much of it is at risk?

Online retailer, December 2009 to December 2011 | £19.64M revenue, 39,516 orders, 5,852 identified customers | December 2011 is a partial month | Guest checkouts (13.1% of revenue, £2.58M) have no customer ID and are excluded from customer-level figures.

The bottom line

- **Repeat buyers are the business.** 72% of customers reorder and generate 97% of revenue, but 51% of identified customers have not bought in 90+ days (£3.30M of past spend).
- **The cheapest lever is the gap between first and second order.** The median is 57 days and only 31% of repeat customers return within 30 days.
- **Revenue is concentrated** in one country (UK, 85.5%), one season (Sep–Nov, 29%) and a few accounts (42 customers, 27%).
- **Recommendation:** start with actions 1 to 3 below. They target the largest revenue at stake and are the easiest to run. All are hypotheses and none has been tested.

Recommended actions, ranked

Ordered by revenue at stake and ease of execution (analyst judgement). Figures are past spend, not forecasts.

#	Action	Who and how much is at stake	Metric to track
1	Follow up new customers at day 30, with a reminder before day 57	All first-time buyers. 27.6% never reorder (23.8% once recent first orders are excluded). A repeat customer is worth about £3,900 in lifetime revenue against £344 for a one-time buyer.	Second-order conversion within 60 days
2	Assign dedicated account management to the 42 top accounts	38 Potential Champions and 4 Wholesale/B2B accounts: £4.61M, 27% of identified revenue.	Order frequency and spend of these 42 accounts
3	Win back lapsed customers, highest value first	Top 100 churned Core/Average customers (£0.88M past spend), then 589 customers lapsed 91 to 180 days (£1.01M). Pre-emptive track for 1,304 active customers with a risk score of 50%+ (£1.06M).	Reactivation rate against a held-out group
4	Add a loyalty or replenishment nudge in months 2 to 3, ahead of Sep–Nov	Retention falls from 21.9% (month 2) to 17.9% (month 6). Target: about 21% at month 6.	Month-6 retention by cohort
5	Reduce dependence on the UK and on Q4	Expand Germany and France, assign account management to the Netherlands and EIRE, run Jan–Aug promotions. A 10% lift on Jan–Aug sales (about £0.65M a month) is worth about £0.5M.	Non-UK share; Jan–Aug monthly revenue
6	Audit the long tail (lower priority)	The bottom 2,344 products generate 4.6% of revenue (£0.91M): candidates to bundle or discontinue.	Revenue kept after bundling or discontinuing

What the data shows (by dashboard page)

- **Revenue concentration.** The UK generates £16.80M (85.5%); 38 of the other 42 countries are below 1% each. Sep–Nov is 29% of Jan–Nov revenue in both years, and growth is flat (+3.0%).
- **Catalog.** 1,028 of 4,688 products (22%) generate 80% of revenue. The top 10 products are only 8.0%, so there is no single-product dependency.
- **Repeat buying.** 4,234 repeat customers (72.4%) generate £16.51M (96.7%); 1,618 one-time buyers generate £0.56M. Time to second order: median 57 days, 31% within 30, 63% within 90.
- **Segments.** Core/Average: 3,830 customers, 65% of revenue. The 42 top accounts hold £4.61M. The Lost segment is 1,980 customers (34%) but only 8.4% of revenue; 368 of them spent over £1,000 each (£0.86M).
- **Retention.** 21.0% in month 1, 21.9% in month 2, 17.9% in month 6, about 15% in months 9 to 10. Returns are higher in Oct and Nov (24% and 28% of a cohort) than in Jan and Feb (13%).
- **Churn.** 2,967 customers (51%) hold £3.30M (17% of revenue): Core/Average £1.76M (53%), Lost £1.44M (43%). Core/Average holds more revenue at risk than the Lost segment.
- **Checks added later.** The repeat-spend gap, the 57-day median and the autumn effect are far larger than chance (Mann-Whitney and bootstrap tests). A lifetime value model puts the next-12-month spend of existing customers at £9.1M, and only 4.7% of it in the Lost segment.

Confidence and limits

- **Descriptive, not causal.** The findings show what happened; the actions are untested hypotheses.
- **Churn** means no purchase for 90+ days, a fixed threshold that can mislabel seasonal buyers: moving it from 60 to 180 days moves the revenue at risk between £4.27M and £2.29M, and about a quarter of the customers labelled churned at 90 days order again within 90 days. “Revenue at risk” is past spend, not expected loss.
- **The risk score** comes from a snapshot of order frequency and spend (ROC-AUC 0.77). A forward-looking version tested on a later period reaches 0.79 to 0.80, but its probabilities over-predict churn in a busier season, so use it to rank customers, not as a forecast of each one.
- **Two definitions of inactive.** “Lost” (1,980 customers) is a K-Means segment; “churned” (2,967) is the 90-day rule.
- **Retention averages** are unweighted across cohorts. Later months include only older cohorts, and the first cohort (Dec 2009) retains 33–50%, so read month 6+ values as indicative.
- **No cost data,** so this is revenue, not profit. Segment names interpret clusters: “Potential Champions” are 38 high-value, recently active customers.

Data: Online Retail II (UCI Machine Learning Repository, CC BY 4.0). Definitions, method and the six-page Power BI report are in the project repository.